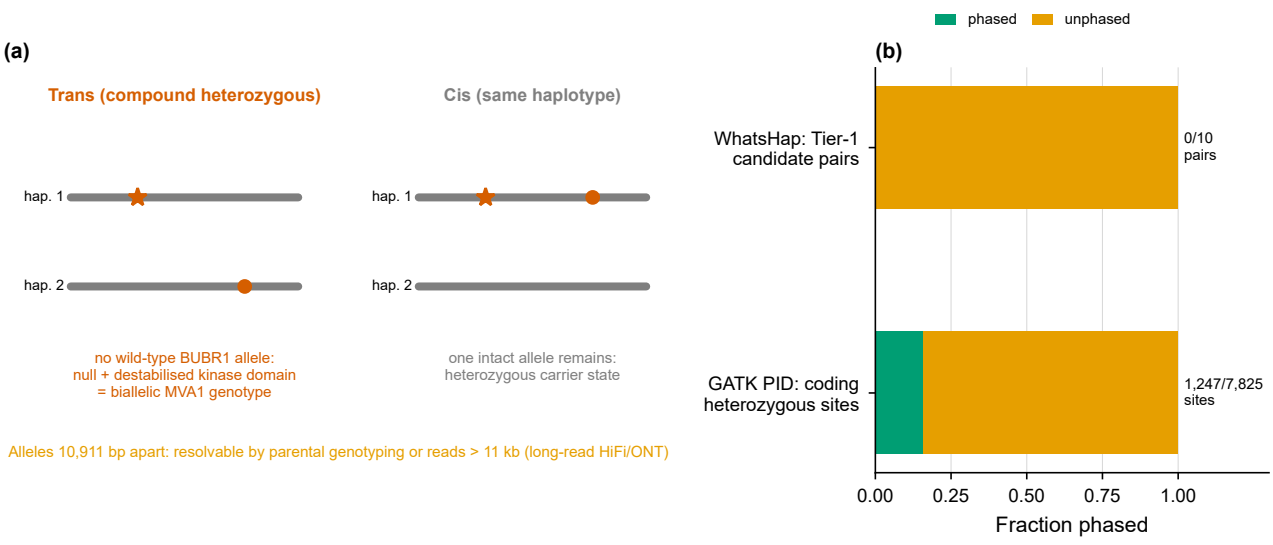
(a)
Trans (compound heterozygous)
Cis (same haplotype)
hap. 1
hap. 2
hap. 1
hap. 2
no wild-type BUBR1 allele:
null + destabilised kinase domain
= biallelic MVA1 genotype
one intact allele remains:
heterozygous carrier state
Alleles 10,911 bp apart: resolvable by parental genotyping or reads > 11 kb (long-read HiFi/ONT)
phased
unphased
(b)
WhatsHap: Tier-1
candidate pairs
0/10
pairs
GATK PID: coding
heterozygous sites
1,247/7,825
sites
0.00
0.25
0.50
0.75
1.00
Fraction phased